EpiProfile_PLANTS — Methodological Whitepaper

Adapting EpiProfile 2.0 for plant histone PTM quantification: a systematic, MSA-driven approach

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1. Motivation

Histone post-translational modification (PTM) profiling by mass spectrometry is well-established in human and mouse chromatin biology, largely thanks to dedicated quantification tools such as EpiProfile (Yuan et al., 2015; 2018). However, plant histones carry amino acid substitutions relative to the human reference that make the original peptide panels inapplicable without modification. Specific residues that serve as PTM sites or propionylation targets differ between species, and new peptide sequences appear that have no counterpart in the human panel.

EpiProfile_PLANTS addresses this gap by extending EpiProfile 2.0 basic to support plant species through a tiered, auditable adaptation process. The methodology is designed so that (a) every change relative to the upstream codebase is classified and traceable, (b) new species can be added systematically by repeating the same workflow, and (c) the resulting bundles are standalone MATLAB packages that a reviewer or collaborator can run end-to-end.

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2. Upstream baseline: EpiProfile 2.0 basic

The starting point is the EpiProfile 2.0 basic codebase by Yuan et al. (publicly available on GitHub). This MATLAB pipeline:

1. Loads MS1/MS2 extracted data from LC-MS/MS runs.
2. Uses a library of histone-derived peptide (hDP) "layouts" — fixed combinations of sequence, charge, and modification mass — to define XIC extraction targets.
3. Builds a retention-time (RT) reference from the top-N most intense runs (via `DrawISOProfile0`), storing it in `0_ref_info.mat`.
4. Quantifies each peptide across all runs by extracting ion chromatograms within mass-tolerance and RT windows (`DrawISOProfile1`).
5. Aggregates results into cohort-level ratio tables (`OutputTogether`, `OutputSinglePTMs`) and optional QC figures (`OutputFigures`).

The human peptide catalog is hard-coded in `init_histone0.m`, which registers each expected peptide with its sequence, m/z values, RT hints, and the quantification module responsible for it.

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3. The tier system (T1-T4)

To maintain full traceability between the plant adaptation and the upstream codebase, every file in a species bundle is assigned a provenance tier:

Tier	Meaning	Example
T1	Identical to upstream — copied without modification	`Getaamass.m`, `find_pair.m`, `H3_01_3_8.m`
T2	Copied from upstream and modified	`init_histone0.m`, `get_rts.m`, `DrawISOProfile1.m`

Tier	Meaning	Example
T3	New — created specifically for plants	`H3_11_3_8.m`, `H3_14_27_40.m`, `H3_Snapshot.m`
T4	Upstream-only — not ported (out of scope)	`Extract_SILAC.m`, `GetaamassH.m`, `DrawISOProfile3.m`

This classification is recorded in `metadata/audit_master.tsv` and described in `docs/tiers.md`. The tier assignment is the primary mechanism by which a reviewer can assess how much of the pipeline is novel vs. reused.

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4. Adaptation methodology (step by step)

Step 0 — Sequence retrieval and Multiple Sequence Alignment (MSA)

For each target species, retrieve the full protein sequences for histone H3 (all variants: H3.1, H3.3, and any species-specific isoforms) and H4 from UniProt or Phytozome.

Perform a Multiple Sequence Alignment using ClustalW, MUSCLE, or equivalent, aligning each plant histone against the human H3.1 reference (UniProt: P68431) and H4 reference (UniProt: P62805).

The MSA reveals:

- **Conserved regions**: peptides identical to human after propionylation + trypsin digestion. These are handled by existing T1 modules.
- **Substituted regions**: peptides where one or more residues differ. These require new T3 modules or T2 modifications.
- **Insertions/deletions**: rare for core histones but checked systematically.

Step 1 — In-silico digestion and peptide catalog

Apply the propionylation + trypsin (Arg-C-like) digestion rules to each aligned histone sequence:

1. Propionylate all unmodified and monomethylated lysines.
2. Cleave at the C-terminal side of arginine (R) residues.

3. Predict the resulting tryptic peptides and their masses.

For each peptide region (e.g., residues 3-8, 9-17, 27-40), compare the plant-specific sequence(s) against the human reference:

- If identical: the upstream quantification module (T1) applies directly.
- If different: a new module is needed (T3), or the existing module must be modified (T2) to handle the variant mass ladder.

Record the full peptide catalog in `init_histone0.m`, including:

- `new_seq`: propionylated peptide sequence
- `new_mz`: monoisotopic m/z at the expected charge state
- `seq_godel`: deterministic hash for RT reference lookup, calculated as `sum((new_seq-'0'+49).*log(2:1+length(new_seq)))`

Step 2 — Writing new quantification modules (TIER3)

Each T3 module follows the upstream H3_XX pattern. The anatomy of a module:

```
function H3_XX_start_end(MS1_index, MS1_peaks, MS2_index, MS2_peaks, ...
                         ptol, cur_outpath, special)
```

Inside the module:

1. **Define the peptide panel**: list all peptideforms (hPFs) for this region — unmodified + every combination of PTMs (me1, me2, me3, ac, etc.) applicable to the sequence.

2. **Calculate m/z for each hPF**: using `Getaamass.m` and `calculate_pepmz.m` to compute the monoisotopic mass, then dividing by charge state.

3. **Retrieve RT reference**: call `get_rts.m` to look up the expected RT from `0_ref_info.mat` (if available) using `seq_godel`.

4. **Extract XICs from MS1**: for each hPF, extract the ion chromatogram within +/- `ptol` ppm around the expected m/z.

5. **Quantify**: call `get_histone0.m` (for unmodified) or the appropriate `get_histoneXX.m` helper to integrate areas and pick the best RT window.

6. **Relocate (if needed)**: for modified forms where signal may be weak, use `relocateD.m` to shift the search window relative to the unmodified peptide's RT.

7. **Validate with MS2**: use MS2 fragmentation data to confirm peptide identity when available (DDA mode).

8. **Write results**: save per-hPF areas, ratios, and RT to a `.mat` file in `cur_outpath/detail/`.

For plant-specific modules, the key difference is the mass ladder: each substituted amino acid changes the peptide mass, the isotope envelope, and potentially the fragmentation pattern. The module must encode the correct mass and any new PTM combinations that become possible (e.g., a K->R substitution removes an acetylation site).

Step 3 — Modifying existing modules (TIER2)

Some upstream functions require changes beyond what a new T3 module provides:

- **`init_histone0.m`**: Expand the peptide catalog to include all plant-specific sequences. This is the central registry where every hDP is defined with its sequence, m/z, charge, and module assignment.

- **`DrawISOProfile1.m`**: The main quantification runner. Must call all T1 modules, all T2 modules, and all T3 modules in sequence. In the upstream codebase, this function was missing from the basic distribution and had to be reconstructed.

- **`get_rts.m`**: Add robustness guards for cases where the RT reference is empty or the peptide is not found (common when first running on a new species without prior RT data).

- **`get_histone0.m`**: Add gradient-length awareness and guards against empty MS1/MS2 arrays that can occur with plant samples of lower abundance than human cell-line extracts.

- **`draw_layout.m`**: Fix XIC plotting for edge cases where no signal is found (prevents MATLAB crashes on empty arrays).

Step 4 — Assembling the species bundle

A complete bundle for species XX is a self-contained MATLAB directory:

```
bundles/XX/src/
  TIER1/    72 files (unchanged upstream)
  TIER2/    ~10 files (modified)
  TIER3/    variable (one per variant peptide region + snapshot aggregator)
```

The user runs the pipeline by calling `EpiProfile.m` from `TIER2/`, which:

1. Calls `init_histone0.m` to register the species-specific peptide catalog.
2. Calls `DrawISOProfile0` to build the RT reference.
3. Calls `DrawISOProfile1` to run quantification.
4. Calls `OutputTogether` + `OutputSinglePTMs` + `OutputFigures`.

Step 5 — Validation

Each bundle is validated against a reference LC-MS/MS dataset:

- **AT (Arabidopsis)**: PRIDE PXD014739 — acid-extracted histones, DDA acquisition on Q Exactive.
- **MP (Marchantia)** and **CR (Chlamydomonas)**: pending validation datasets.

Validation checks:

1. All expected peptides are detected (non-zero area in the ratio table).
2. RT consistency across replicates (CV < 5%).
3. Ratio distributions match biological expectations (e.g., H3K27me3 enrichment in heterochromatin fractions).
4. XIC traces show clean peaks with correct isotope envelopes.

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5. The AT bundle in detail

The Arabidopsis thaliana bundle is the reference implementation. Below is the complete inventory of plant-specific changes.

5.1 TIER3 modules — sequence variants from MSA

Module	Residues	Human reference	AT variants	Substitutions
H3_11_3_8	3-8	TKQTAR	TKQSAR, SNQTAR	T->S (pos 5), TK->SN (pos 3-4)
H3_12_9_17	9-17	KSTGGKAPR	KSTGGKGPR, KSHGGKAPR, ISTGGKAPR	A->G (pos 15), A->H (pos 12), K->I (pos 9)
H3_13_18_26	18-26	KQLATKAAR	KELATKAAR, TLLATKAAR, KQLAPKAAR	Q->E (pos 19), K->T+Q->L (pos 18-19), T->P (pos 22)
H3_14_27_40	27-40	KSAPATGGVKKPHR	KSAPTTGGVKKPHR, QSAPATGGVKKPHR	A->T (pos 31, H3.3), K->Q (pos 27, H3.1-like)
H3_16_53_63	53-63	KYQKSTELLIR	KYQKSTELLNR	I->N (pos 62)
H3_17_73_83	73-83	EIAQDFKTDLR	EIAQDYKTDLR	F->Y (pos 78)
H3_18_117_128	117-128	VTIMPKDIQLAR	VTIMPKDVQLAR, VTIMPKEIQLAR	I->V (pos 123), D->E (pos 123)

Additionally, `H3_06_53_63.m` provides the full PTM panel with relocate logic for the KYQKSTELLIR sequence region (complementing the variant module H3_16_53_63).

5.2 TIER2 modifications summary

File	Change type	Rationale
init_histone0.m	Catalog expansion	56 peptide entries including all AT variants
DrawISOProfile1.m	New (gap-fill)	Main runner calling T1+T2+T3 modules
get_rts.m	Robustness	Guards for empty RT reference

File	Change type	Rationale
get_histone0.m	Robustness	Gradient-aware extraction, empty-array guards
get_histone11.m	Robustness	GetTopBottom11 variant
draw_layout.m	Robustness	XIC plotting fixes
H3_04v3_27_40.m	Variant	KSAPTTGGVKKPHR (H3.3-specific)
OutputFigures.m	Adaptation	QC figures for plant PTM panel
check_otherparas.m	Defaults	ptol=10, ndebug=0
EpiProfile.m	Entry point	Dispatch logic (minimal changes)

6. Extending to new species

The methodology for adding a new species (e.g., MP or CR) follows the same 5-step process:

1. **MSA**: Align the new species' histone sequences against human + AT.
2. **Digest**: Predict propionylation + trypsin peptides.
3. **Compare**: Identify which peptide regions match AT (reuse T3 modules), which match human (reuse T1 modules), and which are unique (write new T3 modules).
4. **Fork**: Copy the AT bundle as a starting point. Update `init_histone0.m` with the new peptide catalog. Add/replace T3 modules as needed.
5. **Validate**: Run against a reference dataset for the new species.

Known differences for planned species:

Species	Key variant (H3 27-40)	Other known changes
M. polymorpha (MP)	KSAPSTGGVKKPHR	H4 20-23: KVFR (L->F)
C. reinhardtii (CR)	KTPATGGVKKPHR	Pending full MSA

7. Data model

The pipeline produces three levels of data:

Level	Abbreviation	Description
Histone-Derived Peptide	hDP	Raw XIC area for a specific peptide + charge + isotope
Histone Peptideform	hPF	A specific modification state of a peptide (e.g., H3K27me3 on KSAPATGGVKKPHR)
Histone PTM summary	hPTM	Site-level PTM abundance (ratio), aggregated across peptideforms

The final cohort export (`histone_ratios.xls`, actually TSV) contains hPF-level ratios, areas, and RTs in a two-header-row format. See `docs/MANUAL.md` section 15 for the exact output contract.

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8. Retention-time reference system

The RT reference mechanism is central to EpiProfile's accuracy:

1. **`DrawISOProfile0`** processes the top-3 most intense runs to establish RT anchors for each peptide, stored in `0_ref_info.mat`.
2. Each anchor is keyed by `seq_godel` — a deterministic hash of the peptide sequence: `sum((new_seq-'0'+49).*log(2:1+length(new_seq)))`.
3. During quantification, `get_rts.m` retrieves the expected RT for each peptide and defines the extraction window.
4. The `relocateD.m` function shifts the search window for modified peptides relative to their unmodified counterpart's RT.

The reference is "global by folder" (`raw_path`), meaning all runs in the same folder share the same RT anchors. This requires that runs within a folder share compatible LC conditions (same column, gradient, instrument).

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9. Quality control

The pipeline includes several QC layers:

1. **XIC visual inspection**: `draw_layout.m` produces per-peptide XIC plots stored in `detail/` for manual review.

2. **Snapshot aggregation**: `H3_Snapshot.m` and `H4_Snapshot.m` compile per-run summaries that can be inspected before cohort aggregation.
3. **OutputFigures**: Generates cohort-level QC plots (bar charts, heatmaps) for rapid assessment of data quality.
4. **Survey template**: `docs/SURVEY.md` provides a structured checklist for documenting each analysis run's parameters and outcomes.

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10. Repository structure

```
epiprofile-plants-main/
  bundles/
    AT/src/TIER1/       72 files — upstream unchanged (T1)
    AT/src/TIER2/       10 files — modified for plants (T2)
    AT/src/TIER3/        9 files — new for AT variants (T3)
    MP/                  planned
    CR/                  planned
  docs/
    MANUAL.md            end-to-end usage guide
    SURVEY.md            structured run-report template
    tiers.md             tier definitions
    provenance.md        file-level provenance tracking
    WHITEPAPER.md        this document
  metadata/
    audit_master.tsv     per-file tier audit (97 entries)
    species.tsv          species bundle registry
  workflows/
    PXD014739/           reference validation dataset (AT)
  CITATION.cff           citation metadata
  LICENSE                GPL-3.0
```

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11. References

- Yuan, Z.F. et al. (2015). EpiProfile quantifies histone peptides with modifications by extracting retention time and intensity in high-resolution mass spectra. *Mol. Cell. Proteomics* 14, 1696-1707.
- Yuan, Z.F. et al. (2018). EpiProfile 2.0: A Computational Platform for Processing Epi-Proteomics Mass Spectrometry Data. *J. Proteome Res.* 17, 2533-2541.
- EpiProfile 2.0 basic source code: upstream GitHub repository.

Document version: 1.0 — Generated 2026-02-18